

RaceDay – API Endpoint Plan

Part 1 – Section B

Organiser – creates and manages events, categories, and results. **Participant** – browses events, enrolls, and views personal results. **Public** – unauthenticated users.

1. Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user as Organiser or Participant. Hashes the password before storing.	Public	{ FirstName, LastName, Email, Password, Role, PhoneNumber }	201 Created – user object (no password). 400 Bad Request – validation errors. 409 Conflict – email already exists.
POST	/api/auth/login	Authenticates a user and creates a server-side session storing UserId and Role.	Public	{ Email, Password }	200 OK – user object and role. 401 Unauthorised – invalid credentials.
POST	/api/auth/logout	Ends the current user session and clears all session data.	Any (logged in)	None	200 OK – session cleared.

2. User Profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/profile	Returns the full profile of the currently logged-in user including extended profile fields.	Any (logged in)	None	200 OK – user and profile object. 401 Unauthorised – not logged in.
PUT	/api/profile	Updates personal information of the currently logged-in user.	Any (logged in)	{ FirstName, LastName, PhoneNumber, DateOfBirth, Gender, EmergencyContact }	200 OK – updated profile. 400 Bad Request – validation errors.
PUT	/api/profile/picture	Uploads a profile picture. File stored in Azure Blob Storage through the API.	Participant	multipart/form-data – file	200 OK – updated ProfilePictureUrl. 400 Bad Request – invalid file.

3. Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Returns all events. Supports filtering by EventType and upcoming dates.	Public	None	200 OK – list of all events.
GET	/api/events/{id}	Returns full details of a single event including its categories.	Public	None	200 OK – event detail. 404 Not Found – event does not exist.
POST	/api/events	Creates a new event. OrganiserId is taken from the session automatically.	Organiser	{ EventName, Description, EventDate, Location, Distance, EventType }	201 Created – event object. 400 Bad Request – validation errors.
PUT	/api/events/{id}	Updates an existing event. Only the Organiser who created it may update.	Organiser	{ EventName, Description, EventDate, Location, Distance, EventType }	200 OK – updated event. 403 Forbidden – not event owner. 404 Not Found.
DELETE	/api/events/{id}	Deletes an event. Only the Organiser who created it may delete.	Organiser	None	200 OK – deleted. 403 Forbidden. 404 Not Found.
PUT	/api/events/{id}/banner	Uploads or replaces a banner image. Stored in Azure Blob Storage.	Organiser	multipart/form-data – file	200 OK – updated BannerImageUrl. 400 Bad Request – invalid file.
GET	/api/events/my	Returns all events created by the logged-in Organiser with total enrolment counts.	Organiser	None	200 OK – list of organiser events.

4. Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Returns all categories for a specific event.	Public	None	200 OK – list of categories. 404 Not Found – event does not exist.
POST	/api/events/{eventId}/categories	Adds a new category to an event. Only the event owner may add categories.	Organiser	{ CategoryName, Description }	201 Created – category object. 403 Forbidden. 404 Not Found.
PUT	/api/events/{eventId}/categories/{id}	Updates an existing category. Only the event owner may update.	Organiser	{ CategoryName, Description }	200 OK – updated category. 403 Forbidden. 404 Not Found.
DELETE	/api/events/{eventId}/categories/{id}	Deletes a category from an event. Only the event owner may delete.	Organiser	None	200 OK – deleted. 403 Forbidden. 404 Not Found.

5. Event Enrolments

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrolments	Enrols the logged-in Participant into an event under a selected category. Status defaults to Pending.	Participant	{ eventId, categoryId }	201 Created – enrolment with status. 400 Bad Request – already enrolled. 404 Not Found – event or category missing.
GET	/api/enrolments/my	Returns all enrolments for the logged-in Participant with event name, category, date, and status.	Participant	None	200 OK – list of participant enrolments.
GET	/api/events/{eventId}/enrolments	Returns all Participants enrolled for a specific event with category and status.	Organiser	None	200 OK – list of enrolments. 403 Forbidden. 404 Not Found.
PUT	/api/enrolments/{id}/status	Updates enrolment status (e.g. Pending to Confirmed).	Organiser	{ Status }	200 OK – updated enrolment. 403 Forbidden. 404 Not Found.
DELETE	/api/enrolments/{id}	Cancels and removes a Participant's own enrolment.	Participant	None	200 OK – deleted. 403 Forbidden. 404 Not Found.

6. Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/results	Captures a finish time and position for an enrolled Participant after an event.	Organiser	{ EnrolmentId, FinishTime, FinishingPosition, TotalFinishers }	201 Created – result object. 400 Bad Request – result already exists. 404 Not Found.
PUT	/api/results/{id}	Updates an existing result record (e.g. to correct a finish time).	Organiser	{ FinishTime, FinishingPosition, TotalFinishers }	200 OK – updated result. 403 Forbidden. 404 Not Found.
GET	/api/results/my	Returns personal race history for the logged-in Participant across all completed events.	Participant	None	200 OK – list of results with event name, date, category, finish time, and position.
GET	/api/events/{eventId}/results	Returns all results for a specific event. Used by Organiser on the results entry page.	Organiser	None	200 OK – list of results with participant names. 403 Forbidden. 404 Not Found.